Quality Flow Edge Validation Pre-Registration (Draft v0.2)

Oct 24, 2026 · @Mickael Lall · Draft v0.2 for independent technical review; changelog in section 16. Not frozen. No new performance test is authorized by this draft.

1. Purpose and status

This protocol defines how we will decide, on untouched evidence, whether Quality Flow has a repeatable advantage, and it is built so the answer can be "no." It is a draft for Muse's independent review; no new performance test is authorized until the protocol is frozen (section 15).

Three questions get three separate verdicts. A PASS on one does not imply a PASS on another.

- **H-BUY:** does the BUY signal predict forward returns beyond a matched baseline?
- **H-EXIT:** do the EXIT rules beat a small set of alternatives on identical entries?
- **H-SYS:** is the complete strategy profitable net of conservative costs?

The claim "Quality Flow has an edge" requires H-SYS to PASS on untouched evidence. H-BUY and H-EXIT explain where an edge comes from or why it is missing.

Three limits, stated up front.

- **Not blind.** The team has seen DEV results, and Claude saw the DEV aggregates in the blind-review packet. Parameters below were set by reasoning about mechanics, not by tuning to DEV outcomes, but their independence from DEV is unproven and Muse should challenge it.
- **Power is limited.** The ledger shows ~1.8R SD per trade, so confirming a +0.17R expectancy needs roughly 700 independent trades. One year of the 43-name universe will not reach that. INCONCLUSIVE is therefore an expected and acceptable outcome, and thresholds may not be loosened after any result is seen (section 6).
- **Scope is the frozen rule set.** We test the rules recorded in experiment QF-EQUITY43-DEV-ORDER-CENSOR-20261023-13 (config pin 87b8285a…8706, source commit 555ebb1) as they are. Any modified rule is a new version and needs fresh untouched evidence; it cannot reuse the holdout.

2. Evidence tiers and firewall

Only untouched evidence can produce a PASS or FAIL; everything else is exploratory and cannot be used to set thresholds, horizons or exits. The packet source already shows development on other windows (a 2022 window and an earlier 4-hour cache), so "untouched" must be proven, not assumed.

Tier	Evidence	Status	Allowed use
0	Windows used in earlier studies (2022 window, earlier 4H cache)	Contaminated	Context only
1	DEV: 2024-09-16 to 2026-10-16, 43 names	Exploratory, results known	Mechanics, variance estimates, descriptive studies (section 14)
2	Sealed holdout: 2023-10-19 to 2024-09-15, same 43 names	Unopened; contamination unaudited	One-shot confirmatory, after section 9
3	New symbols chosen by a pre-declared mechanical rule, never in any prior study	Untouched	Confirmatory on the holdout window; supplementary on the DEV window (shares market days with DEV)
4	Clean forward capture after the freeze date	Not started	Confirmatory, limited by calendar time (section 10)

Firewall rules.

1. Each tier supports one confirmatory test per rule version. A rule change creates a new version that needs fresh evidence; it cannot reuse a tier that was already opened.
2. No result from any tier may change the design of a test on an untouched tier.
3. Holdout access needs two people, a logged session, a frozen code hash, and completed sections 3, 6 and 9.
4. The new-symbol list is generated by a script from a dated source list before any price outcomes are viewed. Muse sets the liquidity cutoff and list source (open item).
5. Every look at Tier 1 is logged in the register (section 13), because each look adds to the multiple-testing burden.

3. Preconditions before any new performance test

No test in sections 5 to 10 runs until every item below is complete and recorded. These fix measurement; none may be decided by comparing performance.

#	Item	Completion evidence
P1	Bar-by-bar mark-to-market drawdown computed and reported next to the event-time figure	Both numbers in the run output; method documented
P2	Ranking rule written down, including whether "contested" uses slots free at the timestamp start or the live count after each fill	Signed definition by Muse and the user, dated before any run
P3	Benchmark window alignment documented as a known limitation and frozen as is	Limitation text in the config
P4	Stop semantics documented as close-evaluated with next-open fill; stale docstring corrected	Code comment fix with new file hash
P5	Cost model converted to per-fill (section 8) and recorded in the config	New config hash
P6	Sizing and exposure rule written down, including peak gross notional versus equity as a reported output	Config entry; peak exposure in output
P7	Unsuppressed signal ledger: every signal bar logged, including those that fall inside a standalone trade or are skipped for gaps	Ledger count exceeds the 590 candidates plus gap skips
P8	Telemetry persisted per signal and trade: full indicator state, exit-decision bar OHLC, per-bar paths, execution records	Fields present in a sample audit
P9	Regression parity: the modified code reproduces the -13 DEV ledger exactly on DEV bars	Byte-identical comparison on the 590 rows

#	Item	Completion evidence
P10	Data audit: bar timestamp semantics, split and dividend adjustment, and session-table coverage checked for every tier to be used	Written audit per tier
P11	Version lock: signed tag recording hashes of code, config, universe list, cost model and session table	Tag and hash list

If any item fails, no test is run, and the failure is recorded in the register.

4. Hypotheses

Each hypothesis has one estimand, one tier of evidence and one verdict. Signal quality is tested without exits (section 5); exits are tested in two ways (section 7): paired on fixed entries, which isolates the exit rule, and in a full-strategy replay, which lets a different exit change later entries. A signal can pass while the exits fail, and the reverse. Every direction below is written so that a positive estimand favors the alternative.

ID	Alternative hypothesis	Estimand (positive favors the alternative)	Evidence	Role
H-BUY-P	The frozen BUY signal beats its matched baseline at 10 bars	Mean paired excess of ATR-normalized forward return, signal minus baseline (section 5)	Tiers 2 to 4	Primary, gating
H-BUY-S5, S20	Same, at 5 and 20 bars	Same	Tiers 2 to 4	Secondary, reported only if H-BUY-P passes
H-EXIT-1	An intrabar stop improves on the close-evaluated stop	Mean paired per-trade net R, E1 minus E0, on fixed entries (study 7A)	Tiers 2 to 4	Gating for the exit verdict

ID	Alternative hypothesis	Estimand (positive favors the alternative)	Evidence	Role
H-EXIT-3	The current exit logic improves on a fixed 20-bar exit	Mean paired per-trade net R, E0 minus E3, on fixed entries (study 7A)	Tiers 2 to 4	Gating: do the exits add value
H-EXIT-4	An intrabar stop improves the complete strategy when later entries are allowed to change	Difference in mean net R per closed trade, replay with E1 minus replay with E0 (study 7C)	Tiers 2 to 4	Gating for any exit change to the deployed system
R-EXIT-2	Robustness, not a hypothesis: how much of E0′s result depends on touch-equals-fill at TP1	Mean paired net R, E2 minus E0 (expected ⩽ 0)	Tiers 2 to 4	Reported with a flag; no PASS or FAIL
H-SYS-1	The complete frozen strategy has positive net expectancy at the gating cost	Mean net R per closed trade at 25 bps round trip	Tiers 2 to 4	Primary for the edge claim

The null for each H is no effect or a negative effect. Direction is pre-declared and one-sided, because only improvement matters for a go decision. H-BUY, H-EXIT and H-SYS are separate families (section 6), and the edge claim rests on H-SYS-1 alone. R-EXIT-2 is outside the Holm family because it is a sensitivity check, not a claim.

5. BUY signal study

The primary endpoint is the mean paired excess of ATR-normalized 10-bar forward return, comparing each signal with a matched baseline drawn from the same symbol. No stops, targets or portfolio rules are involved, so the result says nothing about exits.

Events.

- A signal event is any completed 4H session-aligned bar where the frozen `pine_buy_signal` (V3.6 hybrid: confirmedBuy OR breakoutBuy OR readyBuy) is true after the 215-bar warmup, regardless of portfolio state or any simulated open trade.
- Primary analysis keeps an event only if the same symbol had no signal bar in the previous 20 bars, so 20-bar windows of one symbol never overlap.
- Sensitivity analysis uses every signal bar with cluster-robust inference.
- Events whose forward window runs past the last bar are dropped, never completed with a marked or invented price. Dropped counts are reported.

Forward return. Entry price is the open of the next bar (bar i+1). Exit price is the close of bar i+h. The value is (exit − entry) divided by the signal bar's ATR(14) in price units, which uses only information known at signal time. Percent return is reported alongside.

Primary matched baseline (B-A). For each event, the pool is bars of the same symbol that:

1. start at the same time of day (09:30 or 13:30), so overnight gaps are matched;
2. fall within 30 trading sessions of the event date;
3. have no signal bar within 20 bars on either side;
4. have ATR% (ATR divided by close) within ±25% of the event's;
5. leave a full forward window inside the data.

Draw 20 pool bars at random without replacement (fixed seed). Require at least 8 eligible bars, or drop the event and report the count; more than 15% dropped is a flag. The baseline value is the mean ATR-normalized forward return of the draws, each normalized by its own bar's ATR. The paired excess for an event is its return minus that mean, and the statistic is the equal-weighted mean over events.

Feasibility check, outcome-blind. Before any tier is opened, the matching code is run on DEV in a mode that reports only pool sizes: for every event, the count of eligible bars under rules 1 to 5, with no forward returns computed or shown. If more than 15% of events have fewer than 8 eligible bars, the restrictions are relaxed in a fixed, pre-declared order until the drop rate is under 15%: (a) widen the ATR% band to ±40%; (b) widen the session window to ±45 sessions; (c) lower the minimum to 6 eligible bars. The relaxation step reached is recorded and frozen for every tier. No forward return or group difference is examined during this check, so pool-count feasibility cannot leak outcome information into the design.

This baseline asks whether the signal beats being long the same name at ordinary times with similar volatility. It does not require the signal to beat the trend filter alone.

Diagnostic baseline (B-T, non-gating). The same construction, but the pool is limited to bars where close is above EMA200, EMA21 is above EMA55, and the full signal is false. This isolates what the trigger adds beyond a bullish trend state.

Horizons. The fixed set is 5, 10 and 20 bars, and 10 bars is the single primary endpoint.

- 10 bars is about 5 sessions: long enough to reflect a trend hold, short enough to limit window overlap.
- Disclosure: this choice was made knowing the DEV average hold of about 17 bars, so it is not fully independent of DEV. Muse may argue for 20 before freeze.
- The 5- and 20-bar results are interpreted inferentially only if the 10-bar test passes (fixed-sequence); otherwise they are descriptive.

6. Statistics

Every verdict compares a dependence-aware one-sided 95% confidence bound with a minimum effect δ. PASS means the interval shows, with confidence, an effect of at least δ; it is not enough that the effect is positive and the point estimate looks large. Events are not independent: signals overlap, correlated stocks fire together, and market regimes repeat.

Inference.

- Primary interval: moving-block bootstrap over calendar time, blocks of 20 bars (about 10 sessions), resampling blocks and carrying every event or trade inside each block. Use 10,000 resamples and a seed fixed before any data is opened.
- Robustness interval: a symbol-cluster bootstrap. The verdict uses the wider of the two intervals.
- Always report n, the number of blocks, the design effect (cluster variance divided by independent-observation variance) and n_eff = n / design effect.

Thresholds (proposed; Muse to challenge before freeze).

Test	PASS requires	δ (minimum effect)	Rationale
H-BUY-P	Lower bound L ⩾ δ	0.10 ATR	About 25 bps at a ~2.6% ATR (from the DEV median stop distance of 3.84% = 1.5 ATR; a descriptive figure, not a performance one)

Test	PASS requires	δ (minimum effect)	Rationale
H-EXIT-1, -3, -4	Holm-adjusted lower bound L ⩾ δ	0.10 R per trade	Improvement must exceed a conservative round-trip cost
H-SYS-1	Lower bound L ⩾ δ	0.05 R per trade	About 5 bps of equity per trade at 1% risk, roughly 5% a year at the DEV trade frequency

Reported labels. Alongside the verdict, each test reports one of four labels so a result is never over- or under-stated: CONFIRMED AT δ (L ⩾ δ); POSITIVE, NOT CONFIRMED AT δ (0 < L < δ); INDETERMINATE (L ⩽ 0 and U ⩾ δ); EXCLUDED (U < δ). Only the first maps to PASS and only the last to FAIL. The second is the most likely outcome of a small-sample test with a real but modest effect, and it is reported as exactly that, not as a pass.

Validity floors. Below these the verdict is INCONCLUSIVE regardless of the estimate: 200 de-clustered events for H-BUY, 100 closed trades for H-EXIT and H-SYS, and at least 30 independent blocks and 20 symbols in every test.

Power report, required before any tier is opened. Estimate the outcome SD and design effect from variance only: pool forward returns and baseline returns without computing any group difference, and have an analyst report just the SD and design effect. From these, publish the expected interval half-width for each tier.

- Rough planning figures, to be replaced: about 2.6 ATR SD for the paired 10-bar excess (Claude's estimate) and 1.8R SD per trade (from the ledger).
- At those values, bringing the half-width down to δ takes roughly 1,800 independent events for H-BUY-P and 3,500 independent trades for H-SYS-1, before the design effect. Reaching L ⩾ δ for a true effect of 2δ needs about the same.
- One year of 43 names cannot reach that. A test whose expected half-width exceeds δ is labelled UNDERPOWERED FOR PASS: a PASS is unlikely even if the effect is real. It can still FAIL, because a sufficiently negative result (U < δ) is possible at any sample size, and it can still be INCONCLUSIVE. The label is reported with the verdict. The team decides before opening, not after, whether to add Tier 3 symbols or wait for more forward data.

Multiple testing.

- H-BUY: fixed sequence (10 bars first, then 5 and 20), so no alpha is spent on the secondaries.

- H-EXIT: Holm correction across H-EXIT-1, -3 and -4 at a familywise one-sided 0.05. R-EXIT-2 is a reported sensitivity and takes no alpha.
- H-SYS: a single test at one-sided 0.05.
- The three families answer different questions and are not pooled, and the edge claim rests on H-SYS-1 alone.
- Any hypothesis generated from a DEV look is registered as new and tested only on untouched evidence, with Holm correction across all hypotheses generated from the same DEV exploration.
- The frozen configuration was selected on earlier data from at least 9 ranking variants (rs, vol and composite at top 1 to 3), among other studies. That inflates DEV results but does not invalidate a single frozen test on fresh data.

7. EXIT study

Exits are tested three ways, because a paired comparison on fixed entries and a replay of the whole strategy answer different questions. The arm set is frozen now: no arm may be added or altered after any DEV result is seen, and a new exit idea is a new rule version that needs fresh evidence.

Fidelity to the frozen strategy. Every arm starts from the frozen simulator's own terms: entry at the next bar's open, initial stop = signal-bar close − 1.5 × ATR(14), TP1 = fill + 2.0 × ATR(14), both from the signal bar (`gen_candidates_censored`, matching `pine_backtest.py` lines 150 and 157). Note that the simulator fixes TP1 from signal-bar ATR while the original Pine script recomputes it live; the tests match the rules as frozen and run, not TradingView. Precondition P9 (byte-identical reproduction of the -13 ledger) is the proof that E0 is the strategy and not a look-alike.

Study 7A, paired on event-study entries. Entries are the de-clustered signal events from section 5, run as standalone trades with no ranking, sector cap or position limit. Events that gap below the initial stop are excluded from all arms. This isolates the exit rule on the cleanest, largest entry set, but the entries are not the ones the deployed strategy would take.

Study 7B, paired on the strategy's own candidates (non-gating). The same arms on the 586-style closed candidate set that the frozen strategy generates. Caution: that candidate set is itself produced by E0, because each standalone trade suppresses later same-symbol signals until E0 exits. Comparing other exits on entries that E0 chose is a selection effect in E0's favor, so 7B is descriptive and carries no verdict.

Study 7C, full-strategy replay. The complete frozen pipeline (signal, suppression, ranking, sector cap, position limit, sizing) is re-run with E1 in place of E0, so a different exit changes which later candidates exist and are taken. The estimand is the difference in mean net R per closed trade between the two replays, with trade counts, exposure and drawdown reported beside it. Because the two trade lists differ, inference uses the block bootstrap over calendar time applied to both replays jointly (same blocks, both arms), not a per-trade pairing. H-EXIT-4 is the test; it is the only exit result that speaks to the system as deployed.

Arms. R is always entry minus the initial stop, so arms are comparable.

Arm	Stop	TP1	Other exits	Purpose
E0 (control)	Close below the stop level, exit at next open; runner stop = max(stop, close − 2.5 ATR) after TP1	50% at the limit, filled when the bar's high touches it	Close below EMA55 or trend-bear, next open	The current rules
E1	Intrabar: fill at min(open, stop) with 0.05 ATR extra slippage; stop-first if stop and TP1 are both touched in one bar	Same as E0	Same as E0	Cost and benefit of exiting before the candle closes
E2	Same as E0	Fills only if the high exceeds TP1 by 0.05 ATR	Same as E0	Robustness only (R-EXIT-2): how much E0 depends on touch-equals-fill optimism; expected to lower E0, not improve it
E3 (reference)	None	None	Fixed exit at the close of bar i+20	Do the exits add anything beyond holding the signal

Metric. The gating metric in 7A is the paired per-trade net R difference at the gating cost, written so that a positive value favors the alternative: E1 minus E0 for H-EXIT-1, and E0 minus E3 for H-EXIT-3. R-EXIT-2 reports E2 minus E0 and is expected to be at or below zero; a value below −0.10R raises a FILL-OPTIMISM flag on every E0-based result. Inference uses the block bootstrap and Holm correction from section 6. A paired difference removes entry quality, so signal strength cannot leak into the exit verdict.

Oversized losses (non-gating, always reported).

- Share of trades below −1R, mean excess beyond −1R, expected shortfall of the worst 5% of trades, and the worst-5 sum, for every arm.
- Where persisted bars allow it, split E0's excess loss into the part up to the decision bar's close and the gap from that close to the next open. This is why P8 matters.
- For E1, report how many stops would have recovered above the stop at the candle close. That is the whipsaw cost of exiting early.
- Flag "tail-worse" if an arm improves the mean but its expected shortfall is worse than E0's by more than 0.25R. The flag is printed beside the verdict and does not change it.

TP1 assumptions. The base rule, touch equals a full 50% fill at the limit price, is optimistic because a touch is not a fill and queue position is unknown. E2 tests a conservative rule, and the share of TP1 touches within 0.05 ATR of the level is reported to size the optimism. Real fill behavior comes only from the forward test, which must log orders, acknowledgements and fills.

Evidence. DEV may run 7A, 7B and 7C as an exploratory mechanics check, labelled as such, with no verdict. Confirmatory runs use Tiers 2 to 4 only. Any exit change to the deployed system requires H-EXIT-4 to PASS; a 7A PASS alone does not justify changing the strategy.

8. Costs and execution assumptions

All verdicts use a conservative 25 bps round-trip cost applied per fill, because no real fill record exists for any trade in the record (README missing field 4). The 4 and 10 bps results are reported as sensitivities and never decide a verdict.

- **Per-fill model, stated as an accounting convention.** Cost for each fill = 12.5 bps × (shares filled × fill price), charged in dollars at the fill. Trade net P&L in dollars = Σ(exit fills) − entry fill − Σ(fill costs); net R = that figure divided by risk dollars (shares × (entry − initial stop)). Because exit notional moves with price, total cost is not exactly 25 bps of entry notional: a trade exited 10% higher pays about 26.25 bps of entry

notional, one stopped 4% lower about 24.5 bps. This is intended; it is how a broker charges. The 25 bps figure is the round-trip cost at unchanged price and is the label used in tables.

- **Equivalence note.** Under this convention a path with a 50% TP1 fill is charged 12.5 bps on the entry notional, 6.25 bps on the TP1 notional and 6.25 bps on the final-exit notional, each at its own price.
- **Change from DEV.** The DEV engine deducted one flat fraction of entry price per trade from the blended return (verified in the ledger: 0.0004 and 0.0025). That equals a round-trip charge on entry notional only, independent of exit price and ignoring the TP1 partial. DEV figures are restated under the per-fill convention before any comparison with new results, and both versions are shown once.
- **Fills.** Entries and market exits fill at the stated bar open or close with no partial-fill assumption. TP1 follows the base rule in section 7, stress-tested by E2. E1 stop fills carry 0.05 ATR of slippage on top of the cost model.
- **Not modeled.** Financing, borrow, dividends and taxes. Price adjustment for splits and dividends is audited in P10.
- **H-BUY.** Costs cancel in the paired difference. The report adds a descriptive line, signal excess minus 25 bps, so readers can see whether the effect clears a cost.
- **Sizing.** Trade-level tests are unaffected by sizing. Portfolio-level outputs use the sizing recorded under P6 and report peak gross notional against equity.
- **Fixed in advance.** None of these values may change after any result is seen.

9. Holdout contamination audit

The holdout (2023-10-19 to 2024-09-15) is opened only after this audit finds it clean, and the audit is run by someone who will not run the test. Contamination means that holdout-window outcomes influenced research, rule selection or parameter choice; merely holding historical data that spans the window is not contamination, but it is a reason to trace what was done with it. The record gives reasons to check: it states the holdout content was never parsed, yet the SMCI exclusion cites a holdout evaluability finding, and the source shows earlier studies on a 2022 window and an earlier 4H cache whose dates are not in the packet.

#	Check	Method	Contaminated if
A1	Data lineage	List every cache and archive used in any earlier study (4H cache, 2022 caches, daily 5-year cache, capture archives) with first and last timestamps and file hashes	Not by itself. Overlap means the file is traced through A2 and A4; possession of the bars is not contamination
A2	Access history	Search repository history, issues, notes, notebooks and chat for the holdout label and its dates; collect file access logs for the holdout capture files	A read of holdout outcomes (trades, P&L, aggregates, charts) is found that could have informed research or rule selection
A3	Seal integrity	Recompute holdout file hashes and compare with the committed receipt chains; document exactly what the SMCI/holdout evaluability adjudication inspected	Hashes differ, or bars (not only metadata) were examined
A4	Rule genealogy	For each frozen component (signal constants, exits, ranking, sector cap, benchmark policy, sector map), record the data window and date it was designed or selected on	Any component was designed, selected or adjusted using holdout-window data
A5	Human exposure	Written attestation from each person involved, including Muse, on whether they saw holdout charts, trades, P&L or aggregates	Any exposure other than the legitimate metadata checks in A3

Outcomes, decided jointly by the user and Muse and logged before any access.

- **Clean:** Tier 2 is usable once for the frozen version.
- **Partly contaminated:** Tier 2 is reported as supplementary with no PASS or FAIL standing, and confirmatory weight moves to Tiers 3 and 4.
- **Contaminated:** Tier 2 is retired and the register records why.

Public knowledge of how the 2023 to 2024 market behaved cannot be audited and is disclosed as a residual limitation. Once opened, the holdout is consumed for this rule version: one frozen script, one run, outputs captured, no iteration.

10. Clean forward-test protocol

The earlier forward test (verdict FAIL, recorded 2026-10-03) had known data and timing problems, so it counts as neither support nor rejection here. A new forward test (Tier 4) has two separate dates: a collection start, which can begin as soon as F1 to F5 pass and the version is locked, and a verdict date, which is set by sample size, not by the calendar. Starting clean collection early costs nothing and should not wait for the rest of the protocol to freeze; only the verdict waits.

Gate	Requirement	Pass criterion
F1 Timestamps	One written convention: bar-start stamps in America/New_York, the pinned session table, early closes, and evaluation only after a bar completes. Log capture time against bar end time for every bar.	No signal is evaluated on an incomplete bar; vendor delay measured and documented
F2 Simulator parity	Replay the live-captured bars through the frozen simulator and compare with the live pipeline. Initial check on the first 5 sessions (go/no-go for continuing collection), full check at 20 sessions, then weekly	Signal set identical by symbol and bar time; entry, stop and TP1 within 1 tick; `rs` within 1e-6. Re-run weekly and at the end; any mismatch invalidates the affected period until explained
F3 Telemetry	Persist the P8 fields plus order submit time, acknowledgement, fill time, price and size for every order; archive raw bars as received, with a daily hash	Audit of a sample week shows every field present
F4 Freeze	Signed tag of code, config, universe list, cost model and session table before the first forward bar	Start date is after the tag time

Gate	Requirement	Pass criterion
F5 Operations	No discretionary overrides. Outages and missed signals are logged and counted as unexecuted candidates, never dropped. Use bars as received, not later revisions.	Missed-signal log reconciles to the parity replay
F6 Duration	Collection starts at the version lock. The single verdict analysis happens when the section 6 validity floors are met, projected from candidate counts alone (no performance output), with a minimum of 6 months of collection. At the DEV rate of about 300 candidates a year across 43 names, the 100-closed-trade floor for H-SYS is expected in roughly 6 to 8 months; the H-BUY floor of 200 de-clustered events sooner. Interim looks cover parity and operations only, never performance.	No performance output viewed before the analysis date
F7 Changes	Any change to code, config, universe or cost resets the counter under a new version	Change log is empty, or the version number advanced

Real fills. Simulated and actual fill prices are compared per trade and the slippage distribution is reported. That evidence informs the cost assumptions of future versions. It is not used to rescue this one.

11. Decision rules

Each test returns PASS, FAIL or INCONCLUSIVE from its estimate $\hat{\theta}$, its one-sided 95% interval [L, U] and its minimum effect δ (section 6). The rules are fixed before any untouched data is opened.

Verdict	Condition	Reported label
PASS	Validity floors and all section 3 and section 10 gates met, and L ⩾ δ	CONFIRMED AT δ
FAIL	Floors and gates met, and U < δ. A useful-sized effect is excluded, including a real but too-small positive effect	EXCLUDED
INCONCLUSIVE	Everything else: floors or gates not met, contamination or protocol deviation, 0 < L < δ, or L ⩽ 0 with U ⩾ δ	POSITIVE, NOT CONFIRMED AT δ, or INDETERMINATE, plus UNDERPOWERED FOR PASS where section 6 applies it

Stages.

- **Stage A:** one pooled analysis of Tier 2 plus Tier 3 on the holdout window, once the audit is clean.
- **Stage B:** one analysis of Tier 4 (forward, including forward new symbols), at the date fixed in section 10.
- **"Edge supported"** needs Stage B to PASS for H-SYS-1 and Stage A not to FAIL, because only the forward test has real fills and parity checks.
- **"Edge rejected"** if either stage FAILs H-SYS-1.
- Anything else is INCONCLUSIVE, and it is reported as that.

Reading the three verdicts together.

H-SYS-1	H-BUY-P	H-EXIT-3 (E0 minus E3)	Reading
PASS	PASS	any	Edge supported; the signal contributes, and the exit result shows whether exits add to it
PASS	not PASS	any	Profit is not explained by signal quality; suspect concentration, luck or exit and portfolio effects; unresolved

H-SYS-1	H-BUY-P	H-EXIT-3 (E0 minus E3)	Reading
not PASS	PASS	E0 not better than E3	The signal may carry information that exits or costs consume; any redesign is a new version on fresh evidence
FAIL	FAIL	any	Rejected

No post-hoc changes. δ, α, interval method, block length and cost model may not change after unblinding. A deviation is logged, and the verdict is INCONCLUSIVE unless Muse and the user declared the deviation immaterial before unblinding. Diagnostics in section 12 annotate a verdict and never flip it.

12. Non-gating diagnostics

These describe how fragile or concentrated a result is, and they never decide PASS or FAIL. Trend-following normally depends on a few large winners, so removing the best trades is a diagnostic here, not a requirement. Flags are printed beside the verdict.

Diagnostic	Definition	Flag
Winner concentration	Share of net gain from the top 1, 3, 5 and 5% of trades; mean and profit factor excluding each group	CONCENTRATED if the top 5% of trades supply more than 100% of the net gain
Leave-one-out	Re-estimate dropping each symbol, each sector, and each calendar quarter in turn	FRAGILE if dropping the single best sector or symbol makes $\hat{\theta} \leqslant 0$
Independent periods	Estimate and interval for each calendar half and each tier separately	PERIOD-FRAGILE if fewer than half of the independent periods have $\hat{\theta} > 0$
Market state	Split by whether SPY closes above its 200-bar EMA at the signal bar, a rule fixed now	Reported only
Cost sensitivity	Re-run at 4, 10 and 25 bps round trip	COST-SENSITIVE if $\hat{\theta} \leqslant 0$ at 25 bps but > 0 at 10 bps

Diagnostic	Definition	Flag
Dependence	Design effect, number of simultaneous signals per timestamp, symbols per block	Reported only
Diagnostic baseline	B-T result from section 5	Reported only

No diagnostic may be used to select a subset, filter or parameter for later testing without registering it as a new hypothesis (section 13).

13. Hypothesis register

This register is append-only and keeps every hypothesis and every look at DEV, including failures. Status values are Registered, Run (exploratory), Run (confirmatory), Abandoned and Amended. An amendment is a new row that references the old one, and nothing is deleted. The prior rows below come from the packet source; Muse is asked to backfill dates, windows and results, because the number of earlier variants sets how much to discount DEV results.

ID	Registered	Hypothesis or variant	Tier	Status	Result
H-BUY-P	2026-10-24	Frozen signal beats matched baseline at 10 bars	2 to 4	Registered	—
H-BUY-S5, S20	2026-10-24	Same at 5 and 20 bars	2 to 4	Registered	—
H-EXIT-1	2026-10-24	Intrabar stop (E1) beats close-evaluated stop (E0)	2 to 4	Registered	—
H-EXIT-2	2026-10-24	Conservative TP1 fill rule (E2) vs E0	2 to 4	Registered	—
H-EXIT-3	2026-10-24	E0 beats fixed 20-bar exit (E3)	2 to 4	Registered	—
H-SYS-1	2026-10-24	Complete strategy has positive net expectancy at 25 bps	2 to 4	Registered	—

ID	Registered	Hypothesis or variant	Tier	Status	Result
PRIOR-RANK	Backfill	Nine ranking variants (rs, vol and composite at top 1 to 3); rs top 2 frozen	Prior windows	Run (exploratory)	Backfill
PRIOR-SIGQ	Backfill	Control plus three filters (relative strength, contraction, tight base)	Prior windows	Run (exploratory)	Backfill
PRIOR-EXPO	Backfill	Exposure gates including sector-2, mega-3 and correlation-70 (at least three)	Prior windows	Run (exploratory)	Backfill
PRIOR-STACK	Backfill	Stack cases C0 to C3, including the drawdown gate	Prior windows	Run (exploratory)	Backfill
PRIOR-WIN	Backfill	Studies on the 2022 window and the earlier 4H cache	Tier 0	Run (exploratory)	Backfill
EXP-12	2026-10-23	DEV preflight (package 861f5e2)	1	Run (exploratory)	Stopped at preflight; immutable record
EXP-13	2026-10-23	DEV order and censoring run, frozen C1 configuration	1	Run (exploratory)	4 bps: 187 closed trades, +0.174R, profit factor 1.28; labelled NOT VALIDATED EDGE
FWD-1	Backfill	Earlier forward test	4 (not clean)	Run	Verdict FAIL on 2026-10-03; known data and timing problems; non-confirmatory

14. What DEV may investigate and what needs untouched evidence

DEV can verify mechanics and generate hypotheses. It cannot validate anything, and it cannot decide what the untouched tests look like. Every DEV table carries the heading EXPLORATORY and uses no PASS or FAIL language.

Activity	On DEV?	Needs untouched evidence?
Complete preconditions P1 to P11 and reproduce the -13 ledger	Yes	No
Estimate outcome SD and design effect for the power report (variance only, no group differences)	Yes	No
Regenerate the unsuppressed signal ledger and persist full telemetry	Yes	No
Shake out the event study and matched-baseline code	Yes, exploratory	No
Run exit arms E0 to E3 as a mechanics check	Yes, exploratory; no new arms	No
Explore signal state (ADX, VWAP distance, relative volume and similar)	Yes, as hypothesis generation	Yes, every resulting idea is registered as new and tested only on untouched data
Choose horizons, δ, thresholds, arms and cost model	Fixed at the freeze, before any DEV effect size is examined	Not applicable
Any PASS, FAIL or "edge" statement	No	Yes
Evaluate a modified rule	No	Yes, on evidence that has not been opened for another version
Look at holdout data, even summaries	No	Only through section 9 and a frozen script
Real fills, execution quality and live parity	No	Yes, forward only

Activity	On DEV?	Needs untouched evidence?
Decide which untouched tests to run based on DEV results	Not allowed	Not allowed

15. Open items for Muse and freeze procedure

The protocol freezes only after Muse's independent review and a written resolution of every item below. Review comments should not reference DEV effect sizes.

Open items.

- [ ] Primary horizon: 10 bars as drafted, or 20 bars (section 5).
- [ ] Baseline parameters: ±30 sessions, ±25% ATR, 20 draws, minimum 8 eligible, 15% drop flag.
- [ ] Minimum effects and gating cost: 0.10 ATR, 0.10 R, 0.05 R, 25 bps round trip.
- [ ] Block length (20 bars), the two-interval rule, and the fixed seed.
- [ ] Ranking rule definition (P2): slots free at the timestamp start, or the live count. Which was intended?
- [ ] Sizing and exposure rule (P6), including whether uncapped leverage is intended. Claude's approximation from the ledger puts peak gross notional near 2.8× equity.
- [ ] New-symbol rule (Tier 3): source list, date and liquidity cutoff.
- [ ] Contamination audit (A1 to A5): auditor named, findings, and the overlap between earlier caches and the holdout window.
- [ ] Backfill of the prior rows in the register (section 13).
- [ ] Stage design (A then B) or a single pooled analysis; forward start date and analysis date.
- [ ] Power report result, and the decision to add Tier 3 data or accept that a test is UNDERPOWERED FOR PASS.
- [ ] Any point where Muse and Claude disagree, recorded with each side's reasoning.
- [ ] Matched-baseline feasibility: the outcome-blind pool-count check and the relaxation ladder (section 5).
- [ ] Per-fill cost convention: confirm charging on each fill's own notional is the intended accounting (section 8).

- [ ] Study 7C replay inference: joint block bootstrap over both replays, or an alternative Muse prefers.
- [ ] Forward test: confirm the collection-start versus verdict-date split and the 5-session initial parity check are practical (section 10).
- [ ] Decision labels: confirm $L \geq \delta$ as the only PASS, with POSITIVE, NOT CONFIRMED AT δ reported but never treated as a pass (sections 6 and 11).

Freeze procedure.

1. Muse reviews independently in comments.
2. Disagreements are resolved and logged in a changelog with reasons. The user decides any that remain open.
3. Version 1.0 is committed with a signed tag and hash, and this document is locked.
4. Preconditions P1 to P11 and the power report are completed and published.
5. The feasibility decision (expand the sample, or accept an underpowered test) is logged.
6. The contamination audit (section 9) is completed.
7. Tiers are opened only in the stage order of section 11.

Amendments are allowed only before any untouched data has been opened, and each one is logged. After unblinding there are none.

16. Changelog

Version	Date	Change	Raised by
v0.2	2026-10-24	H-EXIT-3 estimand reversed to E0 minus E3 so a positive value favors the alternative; all estimands now state direction	User review
v0.2	2026-10-24	E2 reclassified as robustness check R-EXIT-2, outside the Holm family, with a FILL-OPTIMISM flag	User review
v0.2	2026-10-24	Cost model written as an explicit per-fill accounting convention; non-equivalence to 25 bps of entry notional stated; DEV restatement required	User review

Version	Date	Change	Raised by
v0.2	2026-10-24	"FAIL-incapable" replaced by UNDERPOWERED FOR PASS; a sufficiently negative result can still FAIL	User review
v0.2	2026-10-24	Outcome-blind feasibility check and fixed relaxation ladder added for the matched baseline	User review
v0.2	2026-10-24	Forward test split into collection start and verdict date; parity check staged at 5 then 20 sessions; duration driven by floors with a 6-month minimum	User review
v0.2	2026-10-24	EXIT study split into 7A (paired, event-study entries), 7B (paired, strategy candidates, non-gating because E0 selects the entries) and 7C (full replay, H-EXIT-4); fidelity to frozen stop and TP1 formulas stated	User review and Claude
v0.2	2026-10-24	PASS tightened to $L \geqslant \delta$; four reported labels added so a positive-but-unconfirmed result is named honestly	User review
v0.2	2026-10-24	Holdout audit: A1 overlap now triggers investigation through A4; contamination requires evidence that outcomes influenced research or rule selection	User review
v0.1	2026-10-24	Initial draft	Claude